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|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11 | A | <p>Above the equator, the gravitational force on the satellite allows it to move in the same direction of rotation as the Earth.</p> <p>At any other latitude, the satellite would either require a constant input of energy to move in the same direction as the Earth or the satellite would be moving in an orbit which changes latitudes as it circles around the Earth.</p> |
| 12 | B | <p>For (monatomic) ideal gases,</p> $U = \frac{3}{2} NkT$ <p><math>N</math> = no. of molecules</p>                                                                                                                                                                                                                                                                               |